

MongoDB Library Exercise

1. Database & Collection Creation

```
mongosh mongod x + v
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C:\Users\Golden Roots>mongosh
Current Mongosh Log ID: 69fb78021e9b766f3e3682d0
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
Using MongoDB:      8.2.7
Using Mongosh:       2.8.2
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-05-01T17:03:45.971+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use library
switched to db library
library> db.createCollection("book")
{ ok: 1 }
library>
```

2. Insert Operations

```
library> db.books.insertOne({
|   title: "Harry Potter",
|   author: "J.K. Rowling",
|   published_year: 1997
| })
{
  acknowledged: true,
  insertedId: ObjectId('69f753a5921b8a1071abc114')
}
library> db.books.insertMany([
| { title: "1984", author: "George Orwell", published_year: 1949 },
| { title: "The Catcher in the Rye", author: "J.D. Salinger", published_year
: 1951 },
| { title: "MongoDB Basics", author: "John Doe", published_year: 2022 },
| { title: "NoSQL Guide", author: "Jane Smith", published_year: 2021 }])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f7549b921b8a1071abc115'),
    '1': ObjectId('69f7549b921b8a1071abc116'),
    '2': ObjectId('69f7549b921b8a1071abc117'),
    '3': ObjectId('69f7549b921b8a1071abc118')
  }
}
library>
```

3. Read Operations

```
}
library> db.books.find()
[
  {
    _id: ObjectId('69f72cedef1b0512d5abc115'),
    title: 'Python Guide',
    author: 'Sara',
    available: true,
    price: 700,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f72cedef1b0512d5abc116'),
    title: 'MongoDB Mastery',
    author: 'Alex',
    available: true,
    price: 600,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f753a5921b8a1071abc114'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    published_year: 1997
  },
  {
    _id: ObjectId('69f7549b921b8a1071abc115'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949
  },
  {
    _id: ObjectId('69f7549b921b8a1071abc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 1951
  },
  {
    _id: ObjectId('69f7549b921b8a1071abc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022
  },
  {
    _id: ObjectId('69f7549b921b8a1071abc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021
  }
]
```

```

library> db.books.find({ author: "J.K. Rowling" })
[
  {
    _id: ObjectId('69f753a5921b8a1071abc114'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    published_year: 1997
  }
]
library> db.books.find().sort({ published_year: 1 }).limit(1)
[
  {
    _id: ObjectId('69f72cedef1b0512d5abc115'),
    title: 'Python Guide',
    author: 'Sara',
    available: true,
    price: 700,
    genre: 'Mystery'
  }
]
library> db.books.updateOne(

```

4. Update Operations

```

library> db.books.updateOne(
  { title: "The Catcher in the Rye" },
  { $set: { published_year: 2026 } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> db.books.updateMany(
  {},
  { $set: { genre: "Mystery" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 7,
  modifiedCount: 5,
  upsertedCount: 0
}
library> db.books.updateOne(

```

5. Delete Operations

```
library> db.books.deleteOne({ title: "1984" })
{ acknowledged: true, deletedCount: 1 }
library> db.books.deleteMany({ published_year: { $lt: 2000 } })
{ acknowledged: true, deletedCount: 1 }
```

6. Advanced Query

```
library> db.books.find().sort({ published_year: -1 }).limit(3)
[
  {
    _id: ObjectId('69f7549b921b8a1071abc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2026,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f7549b921b8a1071abc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f7549b921b8a1071abc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library>
```

```
library> db.books.find(
|   title: { $regex: "MongoDB", $options: "i" }
| })
[
  {
    _id: ObjectId('69f72cedf1b0512d5abc116'),
    title: 'MongoDB Mastery',
    author: 'Alex',
    available: true,
    price: 600,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f7549b921b8a1071abc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  }
]
library> db.books.find(
|   title: { $regex: "NoSQL", $options: "i" }
| })
[
  {
    _id: ObjectId('69f7549b921b8a1071abc118'),
    title: 'NoSQL Guide',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library> db.books.find(
```

```

library> db.books.find({
|   $or: [
|     { title: { $regex: "MongoDB", $options: "i" } },
|     { title: { $regex: "NoSQL", $options: "i" } }
|   ]
| })
[
|   {
|     _id: ObjectId('69f72cedf1b0512d5abc116'),
|     title: 'MongoDB Mastery',
|     author: 'Alex',
|     available: true,
|     price: 600,
|     genre: 'Mystery'
|   },
|   {
|     _id: ObjectId('69f7549b921b8a1071abc117'),
|     title: 'MongoDB Basics',
|     author: 'John Doe',
|     published_year: 2022,
|     genre: 'Mystery'
|   },
|   {
|     _id: ObjectId('69f7549b921b8a1071abc118'),
|     title: 'NoSQL Guide',
|     author: 'Jane Smith',
|     published_year: 2021,
|     genre: 'Mystery'
|   }
| ]
library>

```

Conclusion:

This exercise demonstrates MongoDB CRUD operations including create, read, update, delete, and advanced queries using mongosh.